

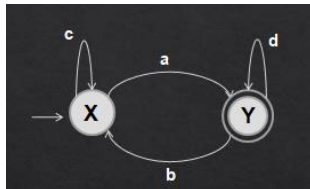
Lecture 11.2: Conversion of DFA to Regular Expression

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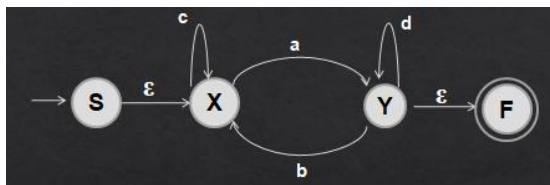
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This is a continuation note of our previous note on conversion of DFA to Regular Expression. Let's see some more examples on which we are going to apply State Elimination Method to convert DFA to Regular Expression.

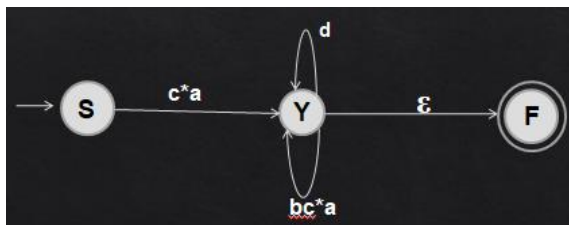
Example 2:



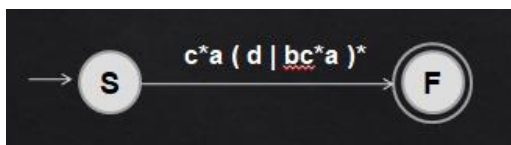
Let's add a new start state and a new final state as per the rules:



Now, let's eliminate state X first (you can eliminate state Y first and X later, doesn't matter)

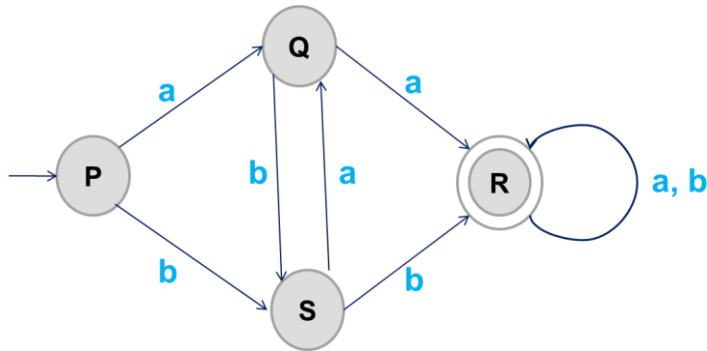


Here, we can see that, the self loop to state X with 'c'-transition is considered and converted as c^* . Now, we need to eliminate state Y.



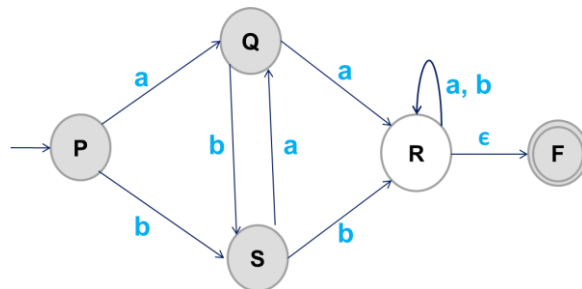
So, the final Regular Expression is: $c^*a (d \mid bc^*a)^*$

Example 3:

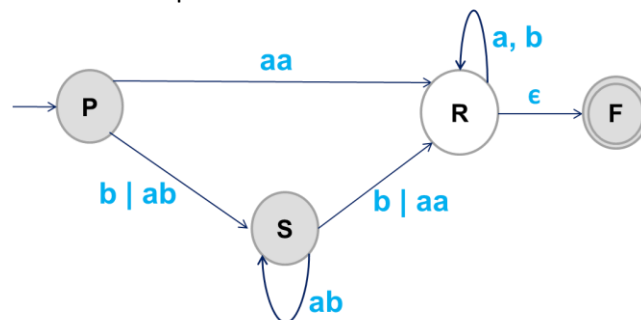


This is the very same example that we got stuck with earlier while figuring out the equivalent regular expression by simply looking at it. Let's convert this DFA to RE in step by step manner:

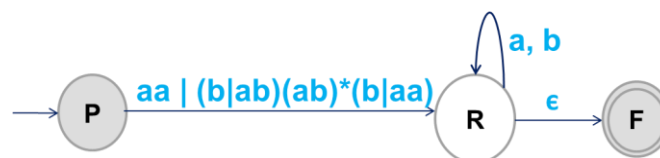
- Step 1: Add new final.



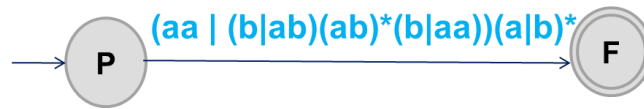
- Eliminate state Q (you can eliminate any state in any order). Update all the transitions which were connected with state Q. For each incoming transition to state Q, consider each outgoing transition from state Q. Following this, $P \rightarrow R$ ($P \rightarrow Q \rightarrow R$), $P \rightarrow S$ ($P \rightarrow Q \rightarrow S$), $S \rightarrow R$ ($S \rightarrow Q \rightarrow R$) and $S \rightarrow S$ ($S \rightarrow Q \rightarrow S$) transitions will be updated.



- Let's eliminate state S next.



- Now, eliminate the only intermediate state R:



So, the equivalent Regular Expression is:

(aa | (b|ab)(ab)*(b|aa)) (a|b)*